



SCHOOL OF COMPUTATION,
INFORMATION AND TECHNOLOGY —
INFORMATICS

TECHNISCHE UNIVERSITÄT MÜNCHEN

Bachelor's Thesis in Informatics

Morphological Inflection Generation

Murilo Escher Pagotto Ronchi





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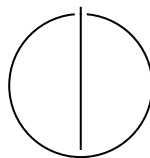
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Morphological Inflection Generation

Titel der Abschlussarbeit

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I confirm that this bachelor's thesis is my own work and I have documented all sources and material used.

Munich, Submission date

Murilo Escher Pagotto Ronchi

Acknowledgments

Abstract

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1 Introduction

Citation test [Lam94].

Acronyms must be added in `main.tex` and are referenced using macros. The first occurrence is automatically replaced with the long version of the acronym, while all subsequent usages use the abbreviation.

E.g. `\ac{TUM}`, `\ac{TUM}` $\Rightarrow$ Technical University of Munich (TUM), TUM

For more details, see the documentation of the `acronym` package¹.

See Table 1.1, Figure 1.1, Figure 1.2, Figure 1.3.

Table 1.1: An example for a simple table.

A	B	C	D
1	2	1	2
2	3	2	3

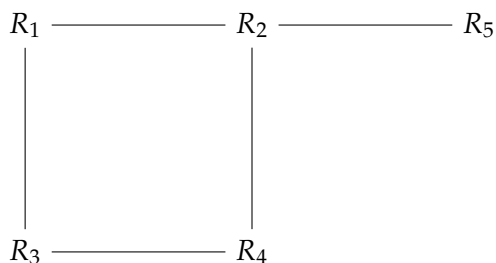


Figure 1.1: An example for a simple drawing.

¹<https://ctan.org/pkg/acronym>

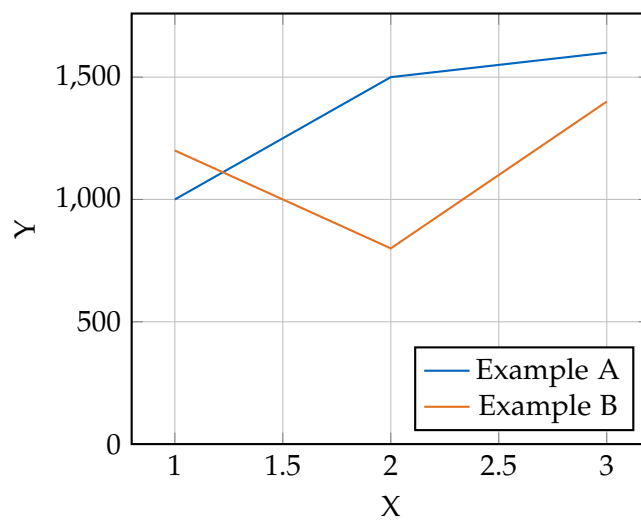


Figure 1.2: An example for a simple plot.

```
SELECT * FROM tbl WHERE tbl.str = "str"
```

Figure 1.3: An example for a source code listing.

2 Background

2.1 Tasks

Different models have been used in this thesis to tackle the problem of morphological inflection generation. This chapter presents the theoretical background of the models used, as well as some related concepts.

2.1.1 Morphological Inflection

Blah inflection

2.1.2 Morphological Analysis

Blah analysis

2.2 Inflection Baselines

2.2.1 Non-Neural Baseline

The first model evaluated was the non-neural baseline provided in the Sigmorphon 2023 shared task [Gol+23] repository. This is a statistical model based on the work first presented in [Cot+17], which uses a set of edit rules extracted from the training data to perform morphological inflection generation.

This rule-based system learns a set of prefix and suffix transformation rules from the training data. For each morphological tag (MSD), the model extracts the possible prefix and suffix transformation rules that map the lemma to the inflected form. During inference, given a lemma and a target MSD, the model applies the most frequent rules associated with the specific transformation to generate the inflected form.

For example, given the Portuguese verb *andar* (to walk) and the target MSD *V;PRS;3;SG* (verb, present tense, third person singular), the model might learn the suffix transformation rule *AR -> A* from the training data. Applying this rule to the lemma *amar* would yield the inflected form *ama*.

Verb	Target MSD	Inflected Form	Transformation Rule
<i>andar</i>	V;PRS;3;SG	<i>anda</i>	AR -> A
<i>amar</i>	V;PRS;3;SG	<i>ama</i>	AR -> A

Table 2.1: Example of transformation rule learned by the non-neural baseline

How the model works, is that it first extracts all possible prefix and suffix transformation rules from the training data. For each lemma-inflected form pair, it identifies the longest common prefix and suffix, and derives the transformation rules by comparing the remaining parts of the lemma and inflected form. The model keeps track of the frequency of each rule for each MSD.

2.2.2 Neural Baseline

Placeholder text.

2.3 ByT5

Placeholder text.

2.4 Large Language Models

Placeholder text.

3 Datasets and Languages

3.1 Datasets

Explain dataset structure, with stem, lemma and inflection. Explain morphological features.

To tackle the task of morphological inflection and analysis, we need labeled data which includes a lemma, the morphological tags and the inflected form. To this purpose, this work used two different, renowned sources: Unimorph CITE and Universal Dependencies CITE.

The data format used for training and evaluating the models followed the one from Unimorph, since it was used in the baselines and thus the start of all experiments. This format consisted of the following:

lemma \t tags \t inflected DO PRETTY SCHEMA HERE

Ex:

EXAMPLE FROM SOME DATASET; ENGLISH FOR SIMPLICITY

3.1.1 Unimorph 4.0

Unimorph is a series of datasets for a variety of languages, comprised of

As mentioned at the start of the section, the data for many of the experiments closely followed the lemma tags inflected format of unimorph.

For languages which were previously tackled by the shared task,

3.1.2 Universal Dependencies

Blah blah UD

3.1.3 Data Preparation

Blah blah conversion between different formats

3.2 Languages

Placeholder text.

4 Methodology

4.1 Morphological Inflection

Placeholder text.

4.1.1 Non-Neural Baseline

Placeholder text.

4.1.2 Neural Baseline

Placeholder text.

4.1.3 ByT5

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4.1.4 Bidirectional Model

Placeholder text.

4.1.5 Evaluation

Placeholder text.

4.2 Morphological Analysis

Placeholder text.

4.2.1 ByT5

Placeholder text.

4.2.2 Bidirectional Model

Placeholder text.

4.2.3 Bidirectional Model with Context

Placeholder text.

4.2.4 Large Language Models

Placeholder text.

4.2.5 Evaluation

Placeholder text.

5 Results

5.1 Morphological Inflection

Placeholder text.

5.2 Morphological Analysis

Placeholder text.

6 Conclusion

Placeholder text.

Abbreviations

TUM Technical University of Munich

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Bibliography

- [Cot+17] R. Cotterell, C. Kirov, J. Sylak-Glassman, G. Walther, E. Vylomova, P. Xia, M. Faruqui, S. Kübler, D. Yarowsky, J. Eisner, and M. Hulden. *CoNLL-SIGMORPHON 2017 Shared Task: Universal Morphological Reinflection in 52 Languages*. 2017. arXiv: 1706.09031 [cs.CL].
- [Gol+23] O. Goldman, K. Batsuren, S. Khalifa, A. Arora, G. Nicolai, R. Tsarfaty, and E. Vylomova. “SIGMORPHON–UniMorph 2023 Shared Task 0: Typologically Diverse Morphological Inflection.” In: *Proceedings of the 20th SIGMORPHON workshop on Computational Research in Phonetics, Phonology, and Morphology*. Ed. by G. Nicolai, E. Chodroff, F. Mailhot, and Ç. Çöltekin. Toronto, Canada: Association for Computational Linguistics, July 2023, pp. 117–125. doi: 10.18653/v1/2023.sigmorphon-1.13.
- [Lam94] L. Lamport. *LaTeX : A Documentation Preparation System User’s Guide and Reference Manual*. Addison-Wesley Professional, 1994.